SOLVING SYSTEMS OF LINEAR EQUATIONS

LINEAR ALGEBRA

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Linear Equation

System of Linear Equations

Solving System of Linear Equations

- By Substitution
- By Elimination

LINEAR EQUATION

$$\boldsymbol{a_1x_1 + a_2x_2 + \cdots + a_nx_n = b}$$

An equation that can be expressed as the **sum of products** of coefficients and variables, such that the sum equals a **constant**

$x_1, x_2, \ldots, x_n$ are the variables

$a_1, a_2, \ldots, a_n$ are the coefficients

b is the constant

LINEAR EQUATION

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

$$2x - 3y + 4z = 45$$

is a linear equation

x, y, z are the variables

$2, -3, 4$ are the coefficients

45 is the constant

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

square of a variable

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

C. $5a + 2 = 6b - \sqrt{2}c$

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

product of variables

C. $5a + 2 = 6b - \sqrt{2}c$

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

C. $5a + 2 = 6b - \sqrt{2}c$

$5a - 6b + \sqrt{2}c = -2$

coeff · variable

SYSTEM OF LINEAR EQUATIONS

SYSTEM OF LINEAR EQUATIONS

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

A **set** of linear equations in the **same variables**

SOLUTION

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

A **set of values** that satisfies every equation in the system

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLUTION

$$\overset{2}{2}(1) + \overset{6}{3}(2) + \overset{3}{3} = 11$$

$$2 + 6 + 3 = 11$$

$$1 + 3(3) = 10$$

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLUTION

$$2(1) + 3(2) + 3 = 11$$

$$2 + 2(3) = 8$$

$$1 + 3(3) = 10$$

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLUTION

$$2(1) + 3(2) + 3 = 11$$

$$2 + 2(3) = 8$$

$$1 + 3(3) = 10$$

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLUTION

$$2(1) + 3(2) + 3 = 11$$

$$2 + 2(3) = 8$$

$$1 + 3(3) = 10$$

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLVING SYSTEM OF LINEAR EQUATIONS

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

$$a - 2b = -10$$

$$a = 2b - 10$$

$$3a + 5b = 14$$

$$3(2b - 10) + 5b = 14$$

$$6b - 30 + 5b = 14$$

$$6b + 5b = 14 + 30$$

$$11b = 44$$

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

$$11b = 44$$

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

$$\frac{\cancel{11}b}{\cancel{11}} = \frac{44}{11}$$

$$\boxed{b = 4}$$

ans

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$a - 8 = -10$$

$$a = -10 + 8$$

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

$$\frac{11b}{11} = \frac{44}{11}$$

$$\boxed{b = 4}$$

ans

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$a - 8 = -10$$

$$a = -10 + 8$$

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$-2 - 2(4) = -10$$

$$3a + 5b = 14$$

$$\frac{11b}{11} = \frac{44}{11}$$

$$\boxed{b = 4}$$

ans

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$a - 8 = -10$$

$$a = -10 + 8$$

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$-2 - 2(4) = -10$$

$$3a + 5b = 14$$

$$\frac{11b}{11} = \frac{44}{11}$$

$$\boxed{b = 4}$$

ans

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$a - 8 = -10$$

$$a = -10 + 8$$

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$-2 - 2(4) = -10$$

$$3(-2) + 5(4) = 14$$

-6 $\quad$ 20

$$\frac{11b}{11} = \frac{44}{11}$$

$$\boxed{b = 4}$$

ans

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$a - 8 = -10$$

$$a = -10 + 8$$

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

eliminate a,

$$-3(a - 2b = -10)$$

$$3a + 5b = 14$$

$$\cancel{-3a} + 6b = 30$$

$$\cancel{3a} + 5b = 14$$

$$\cancel{11}b = 44$$

$$\cancel{11}$$

$$b = 4$$

solution,

$$\boxed{b = 4}$$

ans

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

eliminate $\boldsymbol{b}$,

$$5(a - 2b = -10)$$

$$2(3a + 5b = 14)$$

$$5a - 10b = -50$$

$$6a + 10b = 28$$

$$\frac{11a}{11} = \frac{-22}{11}$$

$$a = -2$$

solution,

$$\boxed{b = 4}$$

ans

$$\boxed{a = -2}$$

ans

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

solution,

$\boxed{b = 4}$

ans

$\boxed{a = -2}$

ans

Check,

$$-2 - 2(4) = -10$$

$$3(-2) + 5(4) = 14$$

Solve the system of linear equations

(1) by **substitution**

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

$$-a + 2b - c = 0$$

$$a = 2b - c$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

$$-a + 2b - c = 0$$

$$a = 2b - c$$

$$2a - 4b - 2c = 12$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$-a + 2b - c = 0$$

$$a = 2b - c$$

$$2a - 4b - 2c = 12$$

$$2(2b - c) - 4b - 2c = 12$$

$$(4b - 2c) - 4b - 2c = 12$$

$$\frac{-4c = 12}{-4}$$

$$c = -3$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b - 2c = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b - 2c = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b - 2(-3) = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b + 6 = 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b \quad = 6 - 6$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$$a = 2b - c$$

$$c = -3$$

$$5a - 5b + 3c = 6$$

$$5(2b - c) - 5b + 3c = 6$$

$$(10b - 5c) - 5b + 3c = 6$$

$$5b \quad = 0$$

$$b = 0$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{a = 3}$ *ans*

$$a = 2b - c$$

$$a = 2(0) - (-3)$$

$$a = 3$$

Solve the system of linear equations

(1) by **substitution**

(2) by elimination

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{a = 3}$ *ans*

Check,

$$2(3) - 4(0) - 2(-3) = 12$$

$$-3 + 2(0) - (-3) = 0$$

$$5(3) - 5(0) + 3(-3) = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b - 2c = 12$$

$$2(-a + 2b - c = 0)$$

$$\begin{array}{r} 2a - 4b - 2c \quad = 12 \\ -2a + 4b - 2c = 0 \\ \hline -4c \ = 12 \\ \hline -4 \end{array}$$

$$c = -3$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b - 2c = 12$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b - 2c = 12$$

$$5a - 5b + 3c = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b - 2(-3) = 12$$

$$5a - 5b + 3(-3) = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b + 6 = 12$$

$$5a - 5b + 3(-3) = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b + 6 = 12$$

$$5a - 5b - 9 = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b \quad = 12 - 6$$

$$5a - 5b - 9 = 6$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$$\boxed{c = -3}$$

ans

eliminate $\boldsymbol{a}$,

$$2a - 4b = 12 - 6$$

$$5a - 5b = 6 + 9$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

eliminate $\boldsymbol{a}$,

$$-5(2a - 4b = 6)$$

$$2(5a - 5b = 15)$$

$$\cancel{-10a} + 20b = -30$$

$$\cancel{10a} - 10b = 30$$

$$10b = 0$$

$$b = 0$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{a = 3}$ *ans*

eliminate $\boldsymbol{b}$,

$$5(2a - 4b = 6)$$

$$-4(5a - 5b = 15)$$

$$10a - 20b = 30$$

$$-20a + 20b = -60$$

$$\frac{-10a}{-10} = \frac{-30}{-10}$$

$$a = 3$$

Solve the system of linear equations

(1) by substitution

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{a = 3}$ *ans*

Check,

$$2(3) - 4(0) - 2(-3) = 12$$

$$-3 + 2(0) - (-3) = 0$$

$$5(3) - 5(0) + 3(-3) = 6$$

SEATWORK